10

Enforcing safety and quality

Canada

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10.1 Introduction

Within the past 50 years there has been a great deal of change in the food industry. The number and variety of food products that are available to today's consumers has increased dramatically and at the same time the consumer's attitude and expectation of food has undergone a significant change. Only a few decades ago, the majority of consumers purchased their fruits, vegetables and meats at the local grocery store and prepared their meals at home. In most cases these food products were produced in local processing plants which received their raw materials from the regional farm and fish industries. These products had limited distribution and were produced in volumes to satisfy local and regional needs.

Now more than ever the consumer is apt to pop something into the microwave, order a pizza or head off to a restaurant for their meals. Technological advances in food processing and the modernization of transportation and food distribution mechanisms has not only increased the variety of the foods we eat but it has also presented new hazards and concerns that must be addressed. Outbreaks of foodborne illnesses now have the potential of being national, continental and even global. At the same time consumers have become more demanding about the content and quality of what they eat, and more aware of food safety issues. All these developments have imposed new demands on government inspection of food manufacturers and retailers.

10.2 The background to food safety inspection in Canada

In Canada, at the federal level, responsibility for food safety and inspection has been shared by four federal departments, Agriculture and Agri-food Canada, Health Canada, Industry Canada and Fisheries and Oceans Canada (Canadian Food Inspection Agency Corporate Business Plan 1998). To enhance the effectiveness and efficiency of the Canadian food inspection system the government of Canada, on April 1, 1997, amalgamated the food inspection activities of these four departments under the Canadian Food Inspection Agency (CFIA), reporting to the Minister of Agriculture and Agri-food Canada.

The CFIA is responsible for all federal food inspection activities extending from the production of animal feeds along the food production chain to the distribution and retail stages. The Agency's mission is safe food, animal health and plant protection. For the first time in the long history of food inspection in Canada, a single federal agency is responsible for food inspection. The Agency's formation brought together 26 different food, animal and plant health inspection programs. These programs share many similar fundamental food inspection principles, goals and objectives but have evolved independently under different environments and different forces. Consequently, different approaches and methods are used in achieving their goals. The level of government intervention and industry responsibility varies from program to program. Products of equal risk, but of different commodity groups, are subject to different inspection regimes. Furthermore, these programs are not static but continue to evolve to meet the changing needs of the marketplace. To prevent continued program divergence and to ensure that program evolution follows a common set of principles and a common discipline, all programs are moving to a common systems-based approach.

10.3 Enforcement: from product to systems inspection

As the food industry has evolved, so has the government's approach to food inspection. The first Canadian Fish Inspection Act was enacted in 1919 to address the fraudulent activities of unscrupulous fish traders which threatened the European and New England markets by damaging the reputation of Canadian fish products. Over time food inspection programs transformed into comprehensive multi-faceted programs that addressed not only fraudulent practices, but also food safety and quality. Government resources were concentrated in the food processing plants which provided an ideal opportunity to inspect the processing conditions and the final products prior to distribution.

These inspections were based on a traditional approach which focused on snapshot inspections of the processing plant environment and comprehensive sampling and inspection of the final product. The plant processing environment was evaluated against prescriptive construction, sanitation and hygiene requirements. The product inspections included label evaluations, organoleptic

analysis and microbiological and chemical analysis. Laboratory analyses are carried out to detect a limited number of chemical, physical and biological indicators of unacceptable products or processes. Under the traditional inspection approach a paternal relationship between the inspector and the processing plant developed where the inspector identified the problems and the plant then took corrective action. In many cases the inspector fulfilled the role of safety and quality control and, because the system relied heavily on final product inspection, mistakes were not identified until they were already processed into the product.

In the 1980s the international marketplace began making greater demands on the food processing industry and government food inspection agencies to provide assurances that food products were nutritious and safe. This was prompted by informed and knowledgeable consumers who now expected zero risk in their food products. As a result, importing countries started to request government certification of more and more food products. Under the traditional inspection program this meant more inspections. However, there were no new resources to perform these additional inspections.

New or re-emerging pathogens have also forced a re-evaluation of previous views and accepted approaches to controlling foodborne illnesses. The emergence of new pathogenic agents such as *E. coli* 0157:H7 has changed the production and process controls for a wide variety of products. Recognized pathogenic micro-organisms are now being identified in new vehicles and products to which they were never linked previously. The pace of change in the types of pathogen risk faced by the food industry meant that traditional inspection regimes, and the operating standards and procedures on which they were based, have struggled to keep up with new hazards. Similarly, the increasing rate of technological and product innovation has also made it more and more difficult to keep official construction, hygiene and operating standards up to date and relevant.

All of these factors have forced industry and governments to look for new innovative ways to achieve safe and wholesome food products. At the same time, government regulatory agencies are also facing new challenges associated with a complex, globalized and technology-driven food industry. Industry has realized the benefits from developing effective food safety and quality control systems and the regulatory agencies have had to develop more efficient techniques to verify the effectiveness of these new industry programs.

Traditional inspection methods, such as sampling and analysing products, or inspecting food processing establishments to assess levels of sanitation and hygiene, are snapshots in time and are reactive to problems already present in the factory or the finished food. This approach does not provide consumers with the desired level of confidence in the food products they consume. The adoption of a systems approach demands that the food industry understands its responsibilities and is knowledgeable of the regulatory and food safety requirements associated with its business. Under a systems approach, industry is required to develop and implement effective control measures to prevent food

safety hazards, fraudulent products, diseased animals or potential plant pests from reaching the marketplace. These control measures are based on science and must be validated as effective. The systems verification approach is preventive and proactive. It moves away from the strategies of 'see a problem – fix it' to a 'see a cause – prevent it' approach. By using such a systems approach, industry can demonstrate that due diligence was practised and product integrity maintained while under its control.

As governments and the food industry move toward a systems approach to food inspection, government's role changes from inspecting for compliance of specific production lots and processing conditions on a specific day, to assessing the effectiveness of industry control measures in achieving food safety and regulatory compliance. Under this approach, government inspectors have many tools to assess the effectiveness of industry's controls. Records of production, control measures and corrective actions, etc. gathered by industry over a period of time can now be reviewed by the inspector. In addition, traditional inspection techniques can be used to focus on areas where non-compliance is identified or simply to verify that the control measures are effective. In effect, the systems approach does not discard relevant traditional inspection methods but continues to use them where it is appropriate. Overseeing by government of industry practices becomes more comprehensive, flexible and responsive to change.

Government food inspection agencies are also responsible for taking the appropriate enforcement action when non-compliance is identified. This can range from detaining a product until further analysis establishes it as either compliant or non-compliant, to the removal of a processor's right to operate. Whatever the enforcement action, it must be fair, predictable and equitable. The systems approach will enhance government food inspection agencies' ability to direct resources based on the level of risk of a product and historical levels of compliance by a food processor.

10.4 HACCP as an industry control system

The development of any management system involves three basic components:

1. Know what standard has to be met.
2. Identify the control measures to be used to meet the standard.
3. Identify the corrective procedures to be followed when the standard is not met.

Once a system is developed, its implementation and maintenance requires a series of support elements to ensure that the system is effective and is being implemented as planned. These elements can be summarised as follows:

- *validation*: before a control system is implemented, the processor must validate the control measures to ensure that they are effective in meeting the required standards

- *control verification*: processors should build into their systems an extra level of controls to verify that the control measures are being implemented as written; verification should include a complete review of the system (for example, annually) as a basis for incorporating changes to products and processes, and making improvements to the system
- *record keeping*: effective record keeping is essential in any control system to provide evidence that the system is being properly implemented; such records will be essential evidence for regulatory authorities in assessing the effectiveness of the control system.

In most respects these HACCP principles correlate closely with the various system elements identified above. One area of difference lies in the contrast between the emphasis in the Codex principles on hazard identification as the first step in a HACCP study and identifying the appropriate standard as the first step in a general systems approach.

Government in Canada has a key role to play in setting standards as a framework within which HACCP planning can take place. These standards also provide the foundation of the food inspection program. This role for government can be accomplished by drafting reference standards for each industry sector which describe the goals and outputs that safety management systems such as HACCP systems must fulfil. These standards, for example, consolidate international requirements in such areas as prerequisite systems (e.g. hygiene procedures), guidance on particular hazards and the setting of critical limits.

10.5 Government food safety programs

The CFIA has several successful programs that are based on the systems approach and provide reference standards for industry to follow.

10.5.1 QMP: fish processors

The CFIA's mandatory Quality Management Program (QMP) applies to all fish, seafood and their products processed for export from Canada or traded interprovincially. Its objective is to verify that these products are processed under conditions that meet all regulatory, trade and food safety requirements. The QMP facilitates the export of Canadian fish and seafood products by meeting international requirements for hazard analysis critical control point (HACCP) systems. The effectiveness of QMPs is verified by CFIA inspectors.

10.5.2 QMPI: fish importers

The Quality Management Program for Importers (QMPI) is carried out by importers who have voluntarily chosen to assume additional responsibility under a shared or enhanced QMPI. In these cases, a written QMPI submission must be

prepared to address licensing and notification, labelling, ingredients, packaging materials, process controls for canned and RTE products, storage, final product and recall procedures. The CFIA reviews and approves the QMPI submissions and conducts verifications of the importer's systems.

10.5.3 Food Safety Enhancement Program (FSEP)

The CFIA's Food Safety Enhancement Program (FSEP) is a voluntary program designed to encourage the development and maintenance of HACCP-based systems in federally registered agri-food processing establishments involved in processing dairy, meat and poultry, processed fruits and vegetables, shell egg and processed egg commodities. Under FSEP, the company is responsible for the development, implementation and maintenance of prerequisite programs (PRPs) and HACCP plans. Written document review and on-site verifications are conducted by the CFIA.

10.6 Regulatory verification versus audit

As government food inspection agencies and the food industry implement food inspection programs based on HACCP systems, the government's role will shift from purely inspection to encompass more of an auditor role. This transition will pose some significant challenges to an organization that has evolved under the traditional approach to inspection. An audit is defined as 'a systematic and independent examination to determine whether quality activities and related results comply with planned arrangements and whether these arrangements are implemented effectively and are suitable to achieve objectives' (ref: ISO 8408). To paraphrase, it is an assessment of a management system, whether it is concerned with quality or safety, to determine whether it is doing what it says it is doing.

The audit takes a systematic approach in that, before the audit commences, the scope is agreed by all parties, the specific elements of the system that will be audited are identified, check-lists are prepared and reviewed by the audit team, specific audit activities are designated to each member of the audit team and the date and duration of the audit is set and agreed by both the auditor and auditee. In order for an audit to be credible it must be performed by an independent and impartial party and the audit results and conclusions must be based on objective evidence that is verifiable. The audit must measure a system against a 'defined' and agreed upon reference standard and be performed by a competent auditor and audit team that is trained and experienced in the sector being audited.

There are two principal activities carried out under the audit. The first is an audit of the company's documented system. This is sometimes referred to as a desk audit as it involves mostly a review of the documented system against the agreed reference standard. The second component of the audit is referred to as the compliance audit. The compliance audit is carried out once it has been

established, through the system audit, that the company's system meets the reference standard. The compliance audit focuses on the application of the system and verifies that the company follows the control procedures as described in their system.

At the conclusion of either the systems audit or the compliance audit, the auditor and the audit team prepares an audit report which identifies the non-conformities or non-compliances that were observed during the audit. The auditee is then required to prepare a corrective action report describing the actions (what, when and who) that will be taken to rectify the non-conformities. The corrective action report is reviewed by the auditor and either accepted or returned to the auditee for amendments. Once corrective actions have been taken, the auditor will determine whether a follow-up visit is necessary to verify whether the relevant corrective actions have been taken and are effective. Once all of the non-conformities have been satisfactorily dealt with by the company, the auditor closes the audit.

The audit approach is a very effective method of testing and challenging a company system such as a HACCP system. An audit may be performed at the request of the company and seen as an opportunity to have an independent and knowledgeable third party assess a HACCP system in a non-adversarial environment. The results of the audit are seen as opportunities to strengthen the system and a necessary step in the cycle of continuous improvement. The relationship between the company and the auditor does not extend outside the mutually agreed scope of the audit.

When comparing the relationship between a government food inspector and a food processing company with the relationship of an auditor and that same company, there are some significant differences. The client of the regulatory inspector is the consumer, whereas the auditor's client is the company. The majority of assessments performed by the government inspector are not at the invitation of the company and at the best of times may be considered by the company to be a distraction and an annoyance. The relationship may be seen by the company as more adversarial since government inspectors are obliged, by the nature of their mandate, to act when a non-compliance with regulations is identified. The auditor passes responsibility for any non-conformity identified in the audit over to the company to deal with. On the other hand, if a government inspector identifies a non-conformity that violates regulatory requirements or may have the potential to generate a health and safety risk to the consumer, the inspector is obligated to take immediate steps to protect the consumer. The government inspector will take the appropriate action depending on the seriousness of the non-conformance but it may require that product be detained or recalled or may possibly involve the suspension of the company's food operation. The government inspector plays an important part in the food control system designed to protect the consumer. The auditor's role, although it has a positive impact, does not carry an equivalent level of responsibility as that of the inspector.

Although the systems approach to assessing industry compliance has required that government inspectors adopt auditing methods and techniques, the

Canadian food inspection system maintains inspectors as the primary assessor of industry compliance. The CFIA has also developed a new approach to assessing industry operating HACCP systems, referred to as 'regulatory verification'.

Regulatory verification applies a combination of audit and inspection techniques in assessing industry compliance and reacting to regulatory non-compliance. This approach provides government inspectors with both audit and inspection tools to assess the effectiveness of industry controls. Auditing techniques such as analysing and verifying documented controls, reviewing records and corrective actions, interviewing company employees carrying out CCP monitoring activities, and observing the application of control activities in-plant, are types of audit techniques used by inspectors when performing a regulatory verification. In addition, traditional inspection techniques can be used to focus on areas where non-compliance is identified or simply to verify that the control measures are effective. The regulatory verification approach is a comprehensive in-depth assessment of industry's controls and outputs which applies both audit and inspection methods. It allows the inspector to evaluate information and data gathered over time by the company, to perform inspections of the product and the plant environment and to focus verification efforts on the key areas for risk and compliance. This allows the decisions of the inspector to be based on a greater amount of information than in the past and gives more flexibility in assessing industry controls.

10.7 Regulatory verification of company food safety systems

The term 'regulatory verification' describes a set of activities that are carried out by, or on behalf of, a government regulatory body, such as the CFIA, to assess compliance of a company's food safety control system (FSCS) to a specified reference standard. The reference standard defines the requirements that the industry FSCS must meet, based on the relevant legislation and regulations. Regulatory verification includes both inspection and audit activities conducted in order to challenge and confirm, or deny, that industry controls are well developed, correctly implemented and effectively maintained.

For businesses that have a documented FSCS, the regulatory verification consists of three distinct phases:

1. The industry self-verification.
2. The assessment of the written system.
3, The assessment of the implemented system.

Regulatory verification is conducted using a team of inspectors under an overall team leader. This does not necessarily mean that a large group of individuals is required to assess an FSCS such as a HACCP system. However, it does reflect the need for a range of talents and a diversity of skills and knowledge. This team can be available in person, or via a network of available experts who can be consulted on particular points or brought in to assist in a particular stage in an

inspection as required. Throughout the regulatory verification process, good communication with industry is emphasized.

10.7.1 Industry self-verification

Prior to implementation of an FSCS, the business should conduct an in-house assessment to determine whether all required components of the system have been addressed. This exercise is called the industry self-verification. In order to assist industry in this step, regulators have developed check-lists based on the appropriate regulatory requirements. Self-verification check-lists should be concise and easy to use. Check-lists have also been used by companies to plan the development of the FSCS as well as a final check before implementation.

The process for industry use is fairly simple. Upon completion of the FSCS, one of the management team, preferably independent of the system development team, should use the check-list to verify that the components are all present prior to submission of the documented plan to the regulatory agency for acceptance. The check-list would then accompany, or precede, the FSCS submission to the regulatory agency.

Industry self-verification can be mistakenly regarded as a 'paper exercise'. This is a false perception. From the food industry's point of view, self-verification check-list is a useful tool for development and completion as well as verification of the FSCS. For regulators, the industry self-verification process helps to bring an FSCS up to the required standard and promotes industry ownership of the FSCS. Check-lists help to determine whether the industry FSCS is sufficiently developed to proceed to the next stage, the systems verification.

10.7.2 Systems verification

The objective of the systems verification is to assess the written FSCS against the reference standard to determine whether the written document is complete and technically and scientifically sound. The process of systems verification has several distinct stages:

- planning the systems verification
- conducting the systems verification
- communicating the systems verification results.

Planning the systems verification

In order to assess and acknowledge any written FSCS, it is essential that a set of criteria be prepared which will serve as the basis for assessment. The criteria must be developed from the reference standard and should interpret the requirements of the reference standard for the written FSCS. The criteria can be developed in the form of a check-list and assessment guidelines to be used by the regulator. Every effort should be made to balance ease of use of the

guidelines with the need for sufficient criteria to conduct a comprehensive assessment of the written FSCS. The systems verification is a well-organized process. Regulators will not begin a systems verification until the industry self-verification indicates that the written FSCS is complete and ready for assessment. It is important to remember that each FSCS is unique, and the regulator must remain open to different approaches in meeting the systems verification criteria.

Conducting the systems verification

The process begins with establishing a systems verification team. This may include both technical or scientific personnel as required. It will be necessary to have all relevant documents available, i.e. the FSCS itself, including any supporting documentation (e.g. the research underpinning hazard analysis), the self-verification report, and any relevant industry guidance documents.

The systems verification is a paper review of all major components of the FSCS. This would include the PRPs and flow diagrams, hazard analysis, control measures, corrective action plans, etc., but would not include very detailed procedures such as specific work instructions. This check-list is used to assess the completeness and soundness of the written FSCS. In making this determination, it is essential that the regulator consider the interaction between various parts of the FSCS, for example the prerequisite systems and the HACCP plan. These elements cannot be assessed independently of one another.

A common difficulty in conducting the systems verification is determining the level of detail required to find the written document complete and sound. It is important to differentiate between 'complete and sound' and the ideal written FSCS that individual regulatory personnel may envisage. The systems verification does not attest to the effective implementation of the FSCS, only that the written system is found to meet the requirement of the reference standard.

Regulators expect that it will take time for a business to create an effective documented FSCS. Time is needed to analyse and develop the first draft of the written system, to test the procedures through implementation, and then to revise the written procedures, at least once but more likely over several system generations. The regulatory system should facilitate, and even encourage, this test-and-revise approach.

Communicating the results

The company should be provided with a written report indicating any deficiencies found in the written FSCS. Regulators are cautioned to avoid the opportunity to suggest ways to revise the written system for two reasons: industry authorship of the document will improve its workability and if regulators are involved in developing the system it may create conflict when the system is being assessed during implementation.

10.7.3 Compliance verification

The compliance verification is the on-site assessment of the implemented FSCS. Compliance verification has a twofold objective: to verify that the FSCS is implemented as written and that the system is effective in meeting the requirements set out in the reference standard.

The compliance verification is based on audit principles. The process is designed to conduct a meaningful assessment in an efficient manner. In order to accomplish this objective, regulators will evaluate a 'thin slice' of the FSCS, i.e. a comprehensive but narrow review of selected aspects of the FSCS. As an example, in a large food production facility a regulator would limit the compliance verification to an in-depth investigation of the controls in place for only one of many products produced.

The compliance verification has four phases:

1. Planning.
2. Opening.
3. Investigating.
4. Closing.

Planning the compliance verification

The scope of the compliance verification is determined by such factors as plant compliance history and the level of risk presented by the product and processes involved. The first step in the compliance verification is the development of the compliance verification check-list. The compliance verification check-list is prepared using the company's FSCS and in advance of the on-site verification. Check-lists are devised to address the scope and objectives of the verification. The check-list should contain specific activities to be conducted to test the application and effectiveness of the FSCS. Appropriate activities include interview questions for key personnel, observation of specific operational procedures, reviewing records, monitoring testing procedures, and the sampling of materials for analysis. The check-list may be expanded during the course of the compliance verification if required in order to determine compliance to the reference standard. The compliance verification check-list is an important record of an individual assessment. With each new assessment of a plant, the previous check-list should be reviewed and used as the starting point for the creation of the new check-list.

Opening the compliance verification

The compliance verification begins with an opening meeting with the plant management. At this meeting, the inspection team should explain the objectives of the compliance verification and its scope. At this time, company representatives should pose any questions of their own regarding the compliance verification. The plant management should be encouraged to provide someone to accompany the regulatory personnel during the compliance verification. When an industry representative accompanies the regulator, a number of benefits are realized:

- the representative witnesses the regulator's observations in real time, and is more likely to understand the reasons behind the inspectors' eventual findings
- the company representative can provide answers to questions immediately
- the company representative can introduce the inspection team to key personnel whom the inspectors may wish to question.

Involving the company in this way improves the transparency of the regulatory process and communication between the two parties. Wrap-up meetings should be held with the processor each day that the compliance verification continues, and so that plant management is kept informed of progress. Of course, when any issue of critical significance is found (i.e. relating to the safety of a food product) the plant management must be informed immediately.

The investigation

To conduct the compliance verification, the regulator follows the plan established by the compliance verification check-list and proceeds with the investigation. The investigation is a series of planned activities to collect objective evidence in support of the inquiry. The results of all findings are recorded. Objective evidence is qualitative or quantitative information. Examples of objective evidence include:

- information contained on company records
- facts related during an interview with a plant employee
- inspector observations
- laboratory results
- product inspection results
- measurements made by an inspector.

An instance of system failure is called a non-compliance or non-conformity. A non-compliance may be either procedural or performance-related and minor or critical with respect to food safety. When a regulator has reason to believe that there is a non-compliance, this belief should be supported by the kind of verifiable evidence noted above.

Closing the compliance verification

The regulator‘s findings, including any non-compliances (supported by evidence), are documented in the compliance verification report and presented to the company at a closing meeting. Since the regulator has a daily wrap-up meeting with the company, the contents of the final report should not be a surprise to the company. The purpose of the closing meeting is to discuss all of the findings of the compliance verification with all relevant levels of the company management to ensure that there is an understanding of the results.

During the closing meeting, the company will be asked to initiate corrective action plans for each non-compliance identified. When the company has provided corrective action plans, at the meeting or at a later date, these will be reviewed and, if satisfactory, will be accepted by the regulator. The regulator

will also verify at a later time that the corrective action is implemented by the company.

10.8 Common barriers to regulatory verification

There are two kinds of barrier to effective regulatory verification:

1. Barriers within regulatory agencies themselves.
2. Barriers within industry.

10.8.1 Barriers within regulatory agencies

Government agencies may encounter difficulties in moving from a traditional inspection approach to a regulatory verification approach. It is important for regulatory agencies to recognize barriers and develop the appropriate strategies to overcome them. Common challenges to instituting regulatory verification include:

- the cost of change
- people.

The cost of change

Over the long term, a regulator verification approach to assessing industry control systems will cost less and provide greater assurance of safety in the food system than a traditional approach. However, the transition period between programs will be a burden on both monetary and human resource. Governments should expect to go through several years of implementation before realizing cost savings. In assessing the cost of introducing a systems approach to food inspection, government should take account of the cost of not moving in this direction.

People

Regulatory personnel may be resistant to change, attributable to moving from the known to the unknown territory. The movement to regulatory verification is accompanied by the need for new skills, such as audit techniques. The attitude of resistance to change may be overcome by communicating the benefits of using a systems approach to food safety, both to the public, the industry and the regulatory personnel.

Regulatory personnel may feel uncomfortable with the new knowledge, skill and expertise requirements associated with an industry system verification approach. Regulators will need a good understanding of the hazards associated with food products, processes and their controls. Regulatory verification relies on the personal judgement of trained, experienced personnel. The approach recognizes human experience, memory, perception and creative thinking as perhaps the most powerful assessment tools available. In contrast to the rigid standards and pass–fail criteria associated with traditional inspection methods, regulatory verification requires a more proactive and flexible approach.

10.8.2 Barriers within industry

There are a number of barriers within industry to the adoption of HACCP systems:

- motivation
- attitudinal change
- experience.

Motivation

In some sectors, such as fish processing, food safety systems based on HACCP principles (such as QMP) are mandatory for businesses trading overseas or between provinces within Canada. This program has covered the great majority of larger and medium-sized businesses in this sector. In common with experience in other countries, wider adoption of HACCP systems such as the voluntary FSEP has been most widespread among larger processors with significant export activity, where adoption of HACCP systems is a necessary step to exporting to the USA or the EU. In these cases the commercial rationale is strong and resources and skills are more readily available to develop HACCP systems.

Small and medium-sized enterprises (SMEs) have been slower to adopt HACCP systems. There has been greater unfamiliarity with the concept, less of a commercial rationale to adopt HACCP systems and concerns about the costs and resource requirements involved. This picture is gradually changing as small businesses become more aware of HACCP systems and their potential benefits, particularly through contact with government inspection staff who are able to explain the real commercial advantages in such areas as enhanced productivity and product quality and increased attractiveness as suppliers.

Attitudinal change

The traditional relationship between inspectors and food processors was a paternalistic one, with inspectors effectively performing a quality control role for some businesses. Processors concentrated on production, fixing safety and quality defects retrospectively as they were identified through inspection of finished products. Implicit in this relationship was the belief that safety was, at least in part, the responsibility of government. This culture of dependence also assumed that companies fixed problems as they arose rather than trying to anticipate and prevent them.

HACCP systems require an active approach to the management of food safety with the emphasis on prevention rather than cure. They require businesses to use official reference standards to identify hazards, control measures and corrective procedures for themselves, and to set up, implement and validate their own tailor-made food safety systems. Some businesses have found it difficult to make this transition. The result has been to rely too heavily on external advice, resulting in poorly designed and understood HACCP systems that fail to reflect the particular conditions of the business.

The move from an attitude of dependence to one of responsibility also needs to feed through all sections of the business, from the managers of the business to those responsible for monitoring CCPs. All those responsible for CCPs, for example, need to be able to understand a critical limit, identify when it has been exceeded, take the appropriate corrective action, record what has occurred and, ideally, suggest remedial action to prevent a recurrence. Part of the compliance verification process is concerned with identifying how far plant employees understand the new responsibilities they have within a HACCP system.

Experience

The development of a food safety system is a significant challenge for any business, and it may take some time before it is working smoothly and effectively. Indeed, in the regulatory verification process the CFIA explicitly recognizes this initial development phase. The three phases of industry self-verification, systems and compliance verification are designed to identify where a business is in the development cycle. Industry self-verification is the first opportunity for a business to test the completeness of its food safety system against an official check-list. Many businesses use the self-verification process to refine and improve their food safety system before they decide to proceed further with the regulatory verification process. Systems verification provides a further opportunity to test and improve the robustness of a company's food safety system, before the final stage of compliance verification and the on-site inspection and final compliance verification report. This approach has the great advantage to industry of allowing businesses to test and improve food safety systems at a number of points, making the development process much more effective. It allows the CFIA both to assist in the process of development and to make the best use of its resources.

10.9 Future trends: the Integrated Inspection System

Presently, the Agency's inspection programs work well in maintaining Canada's food inspection system as among the best in the world. Naturally, these programs will continue to evolve and improve in order to meet the challenges of new hazards, pests and diseases, and to respond to the advancing globalization of trade. The main challenge for the CFIA is one of integrating these programs effectively. The formation of the CFIA has brought together some 14 separate inspection programs. While sharing many of the same principles and goals, each had evolved separately and consequently had differing approaches and methods. The level of government intervention and industry responsibility varied, and products of equal risk, but of different commodity groups, were subject to differing inspection regimes. Finally, individual programs are at differing stages in the move from an inspection to a systems approach.

In order to prevent continued program divergence and to ensure that program evolution follows a common set of principles and a common discipline, the

Integrated Inspection System (IIS) concept was proposed. In the Corporate Business Plan (1997–2000), the CFIA identified the development of the IIS as the mechanism that will guide the evolution of all Agency inspections programs under a consistent approach.

The proposed IIS can be described as one inspection system for all food commodities where industry is responsible for controlling their product and processes in compliance with recognized standards, and government is responsible for setting standards, verifying the effectiveness of industry's control systems and taking appropriate interventions as necessary. The objectives of the proposed IIS are as follows:

- to provide uniform food health inspection strategies that provide an appropriate level of food safety and consumer protection
- to address international regulations in order to facilitate market access for Canadian food, animal, plant and forestry products
- to provide an effective and efficient food inspection system that is open and transparent to all stakeholders
- to integrate the goals, objectives and activities of all players along the food continuum.

Integration under the IIS contains two parts: internal and external.

10.9.1 The internal integration of government inspection programs

The proposed IIS model describes the conceptual framework for development of the IIS. The IIS model contains the IIS Reference Standard and the IIS Verification Reference Standard. The two Reference Standards serve as blueprints to guide the development of industry control systems and the government verification system to assess the effectiveness of industry controls. The reference standard has been based on concepts from the ISO 9000 series and the fundamental principles of audit. It specifies the safety and quality system requirements designed for application by all stakeholders, including industry, government and third parties, along the food continuum.

The Reference Standard contains eight basic elements (see Table 10.1). The application of the Reference Standard will be flexible and not all elements will be applicable to all inspection programs. In some cases the controls may be very basic and focus on maintaining a sanitary environment for handling food products. In others, there may be comprehensive regulatory and trade requirements and specific food safety hazards that must be controlled. Inspection programs will be required to take account of all elements of the Reference Standard.

The proposed IIS Verification Reference Standard will identify the government's strategy to verify industry control systems. The strategies will be scaled appropriately to reflect industry's control measures. For example, traditional inspection technique is a strategy used for industries that do not have any control systems in place. Auditing techniques will be used where safety and

Table 10.1 IIS Reference Standard elements

1. Scope and application
2. Background information
3. Control elements
   - environmental controls
   - hazards controls
     - food safety controls
     - animal health controls
     - plant protection controls
   - regulatory/trade controls
4. Recalls and traceability
5. Training
6. Record control
7. Document control
8. Verification

quality management systems have been implemented. In the instances where a third party is involved in verifications of industry's controls, the CFIA will take on a role similar to an ISO registrar to assess the third party's verification system.

10.9.2 External integration along the food continuum

The second component critical to the development of the IIS is the proposed external integration of food safety and quality strategies along the food continuum. The objective is to examine current food safety control strategies and to determine whether they are the most effective and appropriate. It will provide the opportunity to interlink and build on the different control measures and address any risks that may have been overlooked currently.

It is proposed that this initiative will be conducted with representation from all segments of the food industry from production to retail, consumer groups, federal and provincial food inspection agencies, and academia. The task will be to map out the food continuum for their specific products and then, with scientific support, identify the hazards along the food continuum related to safe food, consumer protection and export market access. The current control measures will then be evaluated to determine their effectiveness and efficiency in preventing, eliminating or reducing the hazards to acceptable levels. In cases where the inspection system can be enhanced, the IIS Reference Standard may be used to develop new control strategies. This approach will identify the most effective points for control along the food continuum, the control measures that are to be implemented, who is best suited to deliver the controls, how the controls will be verified and who is best suited to verify them.

The strength of the integrated food control system will depend on the involvement of all stakeholders in working cooperatively to achieve the desired outcome. The result will be a food control system developed, verified, communicated and implemented by all stakeholders. The benefits are far-

reaching. The industry will be able to build valuable partnerships and to implement more efficient systems to ensure that their products meet all applicable food safety, regulatory and trade requirements. Consumers and our international trade partners will have greater confidence that the products will meet their expectations of safety and consumer protection. Regulatory agencies will be able to make more effective use of their resources to direct their activities corresponding to the level of risk.

10.10 Summary

There are a number of features which, collectively, make the proposed integrated food control approach to HACCP implementation distinctive:

- the focus on setting specific industry sector standards, based on HACCP principles, as a framework within which industry can design tailor-made HACCP systems for each business
- the regulatory verification process itself which makes the best use of audit and inspection skills in ensuring industry compliance with relevant food safety standards
- the stepped approach to regulatory verification which maximizes cooperation between the regulatory authority and industry in developing and improving food safety systems while making the most effective use of resources
- the emphasis on transparency and good communication in the regulatory verification process, which encourages both sides to learn and improve.

In these ways the proposed approach can help to take the regulation of the food industry into the 21st century.